Backpropagation practice sheet (blank): page 1 of 2

x = 1, y = 2, w = 0.5, b = 0, v = 1, c = 0, learning rate 0.1. $z = wx + b$, $h = \tanh z$, $\hat{y} = vh + c$, $J = 0.5\,(\hat{y} - y)^2$

1. Forward pass: z, h, ŷ, J

2. Chain rule: dJ/dŷ, dJ/dc, dJ/dv, dJ/dh, dh/dz, dJ/dz, dJ/dw, dJ/db

Practice sheet (blank): page 2 of 2

3. Simultaneous update of w, b, v, c; new loss

4. Explain the gradient direction

Check your numbers with docs/handwritten_answer_key.md or results/raw/backprop_derivation_values.txt.